

Vector Space:-

A non empty set V is said to be vector space if it satisfies the following condition.

$$\{(x, y) : x, y \in \mathbb{R}\}$$

1: $V_1 + V_2 \in V$ (closure) $\forall V_1, V_2 \in V$

2: $\forall V \in V \exists W$ such that

$$V + W = W + V = 0 \text{ (Additive inverse)}$$

3: $\exists 0 \in V$ such that

4: $\alpha V \in V \quad \forall V \in V$

$$\forall \alpha \in F \text{ (field)}$$

$\{a + ib : a, b \in \mathbb{R}\}$ set of complex no.
field mean real numbers

5: $1 \in F$ such that:

$$1 \cdot V = V$$

:- Non Vector Set :-

$$\{(x, y) : x > 0, y > 0\}$$

$$\{(x, y) : x + y = 1\}$$

$$\{(x, y) : 2025x + 2026y\}$$

Subspace: ~

$$V = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b \in \mathbb{R} \right\}$$

Subspace:

let v be a

vector space:

A subset $w \subset v$ is said to be a subspace of v if w itself is a vector space over the same field.

Condition:- $w \subset v$

$$w_1 + w_2 \in w$$

$$\alpha w \in w$$

for example

$$w_1 = \begin{pmatrix} a_1 & b_1 \\ 1 & 0 \end{pmatrix} + w_2 = \begin{pmatrix} a_2 & b_2 \\ 1 & 0 \end{pmatrix}$$

$$w_1 + w_2 = \begin{pmatrix} a_1 + a_2 & b_1 + b_2 \\ 2 & 0 \end{pmatrix}$$

is not subspace

Dependence ~

let V be a vector space
over the field $\mathbb{R}$. vector V_1, V_2
are said to be dependent
for Example:—

$$V_1 = \alpha V_2 : \alpha \in \mathbb{R}$$

$$\text{if } \alpha V_1 + \beta V_2 = 0$$

$$\text{or } \alpha, \beta \neq 0$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 1 & 0 \end{pmatrix} \text{ so } V_1 = \alpha V_2$$

Independence:—

if $\nexists$ any $\alpha \in \mathbb{F}(\text{field})$
such $V_1 = \alpha V_2$

$$\text{if } \alpha V_1 + \beta V_2 = 0$$

$$\alpha = \beta = 0$$

$\alpha V_1 + \beta V_2 \rightarrow$ linear combination

Q No 8: ~

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \xrightarrow{R_2 - R_1} \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \end{pmatrix}$$

Swap R_2 with R_1

$$A \xrightarrow{R_2 - R_1} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_2 - R_1}$$

$$x_1 + 0x_2 + 2x_3 = 0 \implies x_1 = -2x_3$$

$$0x_1 + 1x_2 - x_3 = 0 \implies x_2 = x_3$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2x_3 \\ x_3 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$$

$$= \left\{ x_3 \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} : x_3 \in \mathbb{R} \right\}$$

$$\left\langle \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\rangle$$

Linearly dependent / Independent:~

V_1, V_2, V_3, V_4
Linearly Independent:~

$$0 = a_1 V_1 + a_2 V_2 + a_3 V_3 + a_4 V_4$$

$$a_1 = 0$$

$$a_2 = 0$$

$$a_3 = 0$$

$$a_4 = 0$$

$$a_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} a_1 \\ -a_1 + a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$a_1 = 0 \implies a_1 = 0$$

$$-a_1 + a_2 = 0$$

$$0 + a_2 = 0$$

$$a_2 = 0$$

So

$$a_1 = a_2 = 0$$

Linearly dependent set

Example:~

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$$

$$R_2 \times 2$$

$$R_2 = 2R_1$$

So this is linearly dependent
on each other.

$$B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 7 \end{pmatrix}$$

It is linearly independent
set

The two set combination are zero
are called linearly independent

Date: _____

Basic

Day: _____

Ques 1

$$ax^3 + bx^2 + cx + d$$

 $P_2(\mathbb{C})$

Polynomial

$$P_2(\mathbb{R}) = \{ax^2 + bx + c; a, b, c \in \mathbb{R}\}$$

$$(a_1 + ib_1)x^2 + (a_2 + ib_2)x + (a_3 + ib_3) \in \mathbb{C}$$

$$z_1x^2 + z_2x + z_3$$

Basis: -

$$\text{let } v = ax^2 + bx + c$$

$$= a(x^2) + b(x) + c(1)$$

$$= 1, x, x^2$$

$$P_2(\mathbb{R}) = \langle 1, x, x^2 \rangle$$

$$\text{Suppose } ax^2 + bx + c = 0$$

Ques 2
Matrix

$$M_{2 \times 2}(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}$$

$$\text{let } v = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$V = \left\langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\rangle$$

let:

$$a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 0$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = 0$$

$$a = b = c = d = 0$$